THE UNLOX ACADEMY
Weekly Assessment

SQL Foundations

Weekly Assessment

DA / DS Track

Total marks: 100 **Questions:** 40

INSTRUCTIONS

Complete all 40 questions.

The assessment uses a fresh dataset called flipcart.products. Run flipcart_setup.sql in MySQL Workbench BEFORE starting the assessment.

Section A (Theory, 8Q × 2.5) – write your answer letter (a/b/c/d) or short response.

Section B (Output Prediction, 8Q × 2.5) – describe what the query returns (row count, values).

Section C (Applied SQL, 24Q × 2.5) – write the actual SQL query. Formatting counts.

You may use Workbench to test your queries. You may NOT use ChatGPT / Copilot / any AI assistant. Any AI-detected answer receives zero for the entire section.

Submit as one .sql file or Word doc. File name: Week3_Assessment_<YourName>.

The Dataset - flipcart.products

A fictional Indian e-commerce catalogue. One table, 30 rows, 10 columns:

Column	Type	Notes
product_id	INT	Primary key, 1-30
product_name	VARCHAR(80)	e.g. 'Sony Bravia 55 TV', 'boAt Airdopes 141'
category	VARCHAR(30)	Electronics, Apparel, Home, Books, Beauty, Sports
brand	VARCHAR(30)	e.g. Sony, Nike, Prestige, Nykaa
price	DECIMAL(10,2)	Range: ₹249 to ₹79,999
stock_quantity	INT	Range 0 to 500. 2 products at 0 (out of stock).
is_active	BOOLEAN	1 discontinued product (is_active = FALSE)
launched_date	DATE	Range: 2017 to 2024
avg_rating	DECIMAL(3,2)	Range: 3.80 to 4.80. 3 products have NULL (new launches)

Column	Type	Notes
discount_pct	INT	0 to 40

Sample rows

```
product_id | product_name                | category    | brand    | price    |
stock | active | rating | disc
-----------|---------------------------|-------------|----------|----------|-----
--|--------|--------|------
     1     | Sony Bravia 55 TV          | Electronics | Sony     | 65000.00 |  12
|  T     | 4.50   |  15
     2     | Apple iPhone 15            | Electronics | Apple    | 79999.00 |  25
|  T     | 4.80   |   5
     6     | OnePlus Nord Buds 2        | Electronics | OnePlus  |  2199.00 | 180
|  T     | NULL   |  25
     8     | Nokia 105                  | Electronics | Nokia    |  1499.00 | 500
|  F     | 3.80   |   0
    22     | Rich Dad Poor Dad          | Books       | Kiyosaki |   299.00 | 180
|  T     | 4.40   |  30
    27     | WOW Skin Vitamin C Serum   | Beauty      | WOW      |   899.00 |   0
|  T     | 4.50   |  40
```

(30 total rows. Data breakdown: Electronics 8, Apparel 6, Home 5, Books 4, Beauty 4, Sports 3.)

Section A - Theory

Choose the correct option (a, b, c, or d) for each question. Write only the letter.

A1. In MySQL, what is the relationship between a schema and a database?

a) Schema is a group of databases
b) Database is a group of schemas
c) Schema and database are the same thing in MySQL
d) Schema is a virtual view of a database

Your answer: _______

A2. Which SQL command family does TRUNCATE TABLE belong to?

a) DML (Data Manipulation Language)
b) DDL (Data Definition Language)
c) DQL (Data Query Language)
d) DCL (Data Control Language)

Your answer: _______

A3. You are designing a table to store product prices for an e-commerce site (e.g. ₹299.50, ₹79,999.00). Which data type is the most appropriate?

a) FLOAT
b) DECIMAL
c) VARCHAR
d) INT

Your answer: _______

A4. You try to INSERT a row into an orders table with a customer_id that doesn't exist in the customers table. There is a foreign key constraint. What happens?

a) The row is inserted with NULL for customer_id
b) The row is inserted and a warning is logged
c) The INSERT fails with a foreign key constraint error
d) The row is inserted but marked as invalid

Your answer: _______

A5. Which of the following statements is TRUE about WHERE vs HAVING?

a) WHERE and HAVING can be used interchangeably
b) WHERE filters individual rows before aggregation; HAVING filters groups after aggregation
c) HAVING is used for row-level filtering; WHERE is used for group filtering

d) HAVING is faster than WHERE

Your answer: ______

A6. What is the result of the query: SELECT * FROM products WHERE avg_rating = NULL;

a) Returns all rows where avg_rating is NULL

b) Returns all rows where avg_rating is NOT NULL

c) Returns 0 rows (no rows match)

d) Throws a syntax error

Your answer: ______

A7. What is the difference between COUNT(*) and COUNT(column_name)?

a) They are identical

b) COUNT(*) counts all rows; COUNT(column_name) counts only non-NULL values in that column

c) COUNT(*) is faster but less accurate

d) COUNT(column_name) returns only distinct values

Your answer: ______

A8. Which command permanently removes an entire table (both structure and data), and CANNOT be undone by ROLLBACK?

a) DELETE FROM table_name

b) TRUNCATE TABLE table_name

c) DROP TABLE table_name

d) REMOVE TABLE table_name

Your answer: ______

Section B - Output Prediction

For each query below, describe what it returns. Give the row count and/or specific values.

B1. What does this query return?

```
SELECT COUNT(*) FROM products WHERE category = 'Electronics';
```

Your answer:

B2. How many rows does this query return?

```
SELECT * FROM products WHERE price BETWEEN 1000 AND 3000;
```

Your answer:

B3. What is the output of this query?

```
SELECT product_name FROM products
WHERE category = 'Books' AND price < 400
ORDER BY price DESC
LIMIT 1;
```

Your answer:

B4. How many rows does this query return?

```
SELECT * FROM products WHERE avg_rating IS NULL;
```

Your answer:

B5. What single value does this query return?

```
SELECT MAX(price) FROM products WHERE category = 'Books';
```

Your answer:

B6. What does this query return?

```
SELECT category, COUNT(*) FROM products
WHERE is_active = TRUE
GROUP BY category
HAVING COUNT(*) > 4;
```

Your answer:

B7. What does the output look like?

```
SELECT product_name,
    CASE
        WHEN price < 500 THEN 'Budget'
        WHEN price < 5000 THEN 'Mid'
        ELSE 'Premium'
    END AS tier
FROM products WHERE category = 'Beauty';
```

Your answer:

B8. What does this query return?

```
SELECT product_name, COALESCE(avg_rating, 0) AS rating
FROM products
WHERE stock_quantity = 0;
```

Your answer:

Section C - Applied SQL

Write the actual SQL query for each question. Formatting counts - use uppercase for keywords and align clauses on new lines for complex queries.

C1 - Basic SELECT + WHERE + ORDER BY

C1. Write a query to display all products with all columns.

Your query:

C2. Write a query that shows the product_name and price of all Books.

Your query:

C3. Write a query to list all products priced above ₹10,000, sorted from highest price to lowest.

Your query:

C4. Write a query to display the top 5 most expensive products in the Electronics category (show product name and price).

Your query:

C2 - Filtering Variations (IN, BETWEEN, LIKE)

C5. Write a query to list all products in either Electronics or Apparel category. Use the IN operator.

Your query:

C6. Write a query to find all products with a price between ₹500 and ₹2,000 (inclusive).

Your query:

C7. Write a query to find all products whose product_name contains the word 'Watch'.

Your query:

C8. Write a query to find all products whose brand starts with the letter 'S'.

Your query:

C3 - DISTINCT + Aggregate Functions

C9. Write a query to list all the unique categories in the products table.

Your query:

C10. Write a query to find the total number of products in the entire catalogue.

Your query:

C11. Write a query to find the average price of all Books.

Your query:

C12. Write a single query that returns the maximum and minimum price across all products.

Your query:

C4 - GROUP BY

C13. Write a query to count how many products are in each category.

Your query:

C14. Write a query to show the total stock quantity for each category.

Your query:

C15. Write a query to show the average price per category, sorted from highest average to lowest.

Your query:

C16. Write a query to show the count of products and the average price for each brand, showing only brands that have more than 1 product.

Your query:

C5 - HAVING + LIMIT

C17. Write a query to find categories that have more than 4 active products.

Your query:

C18. Write a query to find the 3 most expensive products in the entire catalogue.

Your query:

C19. Write a query to find all categories where the average price is above ₹2,000.

Your query:

C6 - NULL Handling

C20. Write a query to list all products where the avg_rating is missing.

Your query:

C21. Write a query to show product_name and rating for all products. If avg_rating is NULL, display the text 'New Launch' instead.

Your query:

C7 - CASE WHEN

C22. Write a query to display each product with a price_tier column classified as 'Budget' if price < ₹1,000, 'Mid' if price < ₹10,000, or 'Premium' if price >= ₹10,000.

Your query:

C23. For each category, write a query that shows the total product count and the count of 'Premium' products (price >= ₹10,000). Use SUM(CASE WHEN...).

Your query:

C24. Write a comprehensive query showing for each category (only categories with at least 3 products): total product count, count of active products, average price, and a category_tier column ('Cheap' if avg < ₹1500, 'Standard' if avg < ₹10000, 'Luxury' if avg >= ₹10000). Sort by average price descending.

Your query:

Submission

- **File name:** Week3_Assessment_<YourName>.sql
- **Upload to:** The Google Form. Deadline 7 days.
- **Integrity:** This is your work only. AI assistance, copy-paste from classmates, or answers from tutorial sites = zero and re-submission required.

All the best.

— The Unlox Academy